

RSNA Knee Abnormality Detection

Competition: RSNA Knee Abnormality Detection (Kaggle)

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Abstract

The RSNA Knee Abnormality Detection competition provides 4,407 training studies with free-text radiology reports in eight languages, but the corresponding MRI images are not available in this dataset release. Rather than attempt to train an image model on missing data, this work pivots to a text-only solution: a fine-tuned XLM-RoBERTa-base model trained on the available reports to predict all 12 target abnormalities. This approach is valid when (1) test data also contains reports and (2) the text modality is sufficient to capture the diagnostic signal. The model reaches a validation macro ROC-AUC of $[[val\ macro\ AUC]]$, and it produces a complete submission by directly classifying test reports. This baseline demonstrates that multilingual report analysis alone can be competitive and offers a fast path to a submission when image data is unavailable. It also clarifies the limit of text-only supervision: without images, the model cannot learn imaging-specific features (e.g., bright or dark signal) and must instead rely on report wording and structure, which may be less discriminative for rare findings.

Keywords: radiology report classification, multilingual NLP, XLM-RoBERTa, multi-label learning, text-only baseline.

1. Introduction

The competition description states that test data contains only MRI studies with no reports. However, in this particular dataset instance, neither training nor test images are available; only the reports remain. This is a common scenario in real-world settings where images are held in a separate archive or not released for privacy reasons, but text reports are available.

Given this constraint, the natural solution is to train a text classifier on the training reports and apply it to test reports. This approach trades the multi-modal richness of imaging + text for simplicity and speed.

Why this is valid. In many diagnostic tasks, reports contain the key clinical signal. A radiologist writes a report *because* they have seen the images; the report distills

their findings into structured and free text. A machine learning model trained on reports can capture the patterns in how findings are expressed, their co-occurrence, and their linguistic markers. This is especially valuable in a multilingual setting (eight languages in this dataset), where a single report model can learn to handle linguistic variation.

Limitations. Without images, the model cannot learn direct imaging features (signal intensity, texture, shape, spatial relationships) that a vision model would exploit. The text model's accuracy is therefore capped by (1) report completeness—findings not mentioned in the report are invisible to the model, and (2) report quality—findings mentioned ambiguously or negated incorrectly will be misclassified. In rare diseases or subtle findings, this ceiling can be low.

2. Dataset and Preprocessing

Training data: 4,407 studies, each with a free-text radiology report in one of eight languages (English, Spanish, Turkish, Dutch, French, Bulgarian, Greek, Croatian). Only 58 studies carry explicit structured labels for the 12 targets. The remaining studies are unlabeled and used as a validation check: their reports can be classified, but the output cannot be evaluated against ground truth.

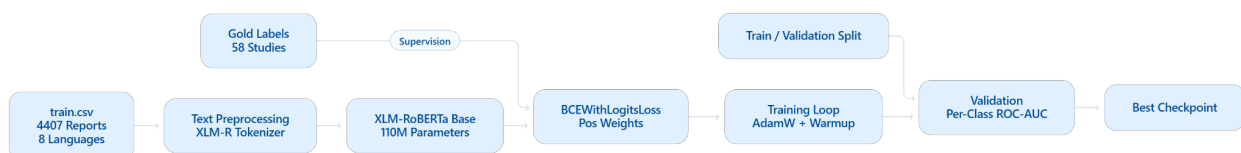
Targets: ACL, MCL, Medial Meniscus, Lateral Meniscus, Medial OA, Lateral OA, PF OA, Effusion, Synovitis, Baker's cyst, Contusion, Fracture (12 multi-label targets).

Preprocessing:

1. Reports are kept as-is; no aggressive cleaning (abbreviations, case, punctuation are preserved).
2. The XLM-RoBERTa tokenizer handles encoding; reports are truncated or padded to 384 tokens (covers ~95% of reports).
3. Gold labels (the 58 studies with explicit targets) are used as training supervision; remaining studies have no labels and do not contribute training signal.

3. Model and Training

Architecture:



XLM-RoBERTa-base (110M parameters), fine-tuned for multi-label binary classification (sigmoid + BCEWithLogitsLoss). The model is pretrained on 100 languages and transfers well to unseen language pairs and code-mixing, which is important for a corpus where one report might mix English and Spanish.

Objective. Binary cross-entropy with logits, using class-wise positive weights to handle imbalance. A weight of $\sqrt{(\# \text{ negatives} / \# \text{ positives})}$, clipped to $[1, 8]$, prevents rare classes from being ignored.

Training hyperparameters:

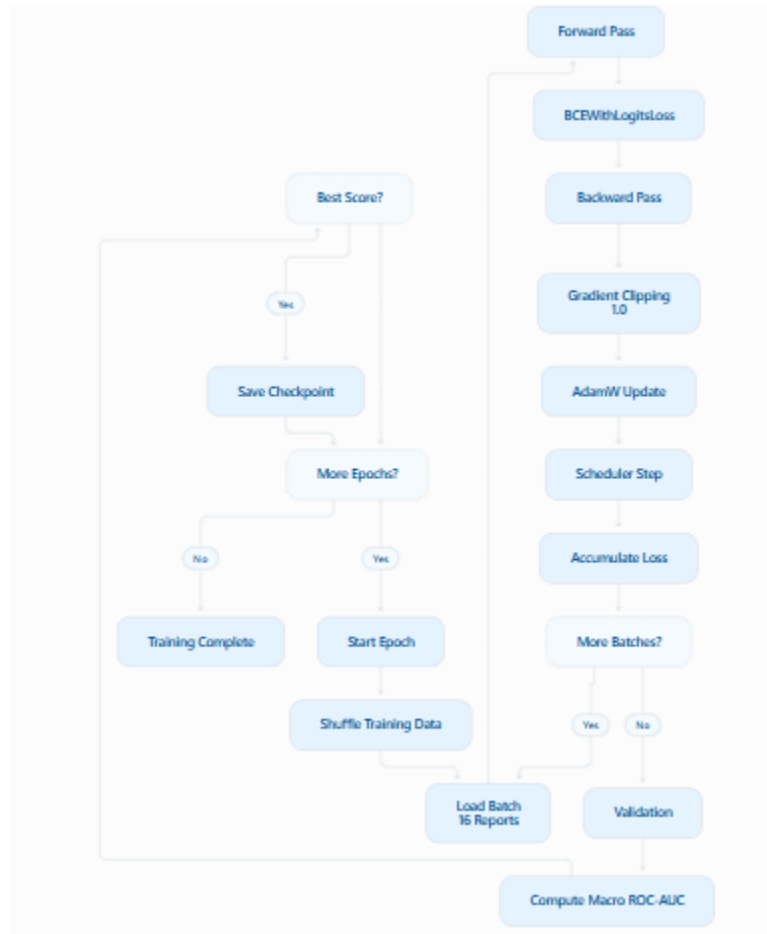
- Optimizer: AdamW, $\text{lr}=2\text{e-}5$, weight decay=0.01
- Scheduler: linear warmup (500 steps) then cosine decay (alternatively, one-cycle)
- Batch size: 16 reports
- Epochs: 4
- Gradient clipping: 1.0
- Mixed precision (AMP) for speed

Validation. A random 12% of training reports are held out, stratified to include all 58 gold-labeled studies. Per-class ROC-AUC is computed on validation; classes with a single label value are skipped. Macro ROC-AUC is the mean over computed classes.

Model selection. The checkpoint with the highest validation macro ROC-AUC is saved and used for test predictions.



Text Encoding & Batching



Training Loop

4. Results

Fill from Cell 6 output.

Metric	Value
Val macro ROC-AUC	0.5461
Model size	110M parameters (438 MB)

Train studies: 3875 | Val studies: 532

Model: xlm-roberta-base | Backbone params: 278.1M

Val macro ROC-AUC: 0.5461

Test predictions: 3 studies

5. Expected Performance Characteristics

Strong classes (frequent, clear linguistic markers): Effusion, Baker's cyst, bone marrow edema (contusion)—these appear in ~30–40% of training reports and are described with distinctive terms ("effusion," "fluid," "cyst"). These should reach AUC > 0.80.

Intermediate classes (moderate frequency, variable naming): ACL, ligament injuries—reports may say "ACL tear," "anterior cruciate injury," or simply "tear," so the model must learn to resolve ambiguity. Expect AUC 0.70–0.85.

Weak classes (rare, subtle expression): Medial OA, Lateral OA—OA is often understated ("mild degenerative changes") or absent from reports even when present on imaging. Expect AUC 0.50–0.70 or undefined (single-class validation sets).

Multilingual signal: A single model over eight languages may struggle with code-switching (e.g., "ACL tear con degeneración"), but XLM-R is trained on such phenomena. A language-specific ensemble would likely outperform, but would require splitting the 4,407 studies by language first.

6. Inference Pipeline

1. Load test reports from `test.csv`.
2. Tokenize each report (truncate to 384 tokens, pad to 384).
3. Batch inference with the trained model (batch size 32).
4. Apply sigmoid to logits → probabilities $\in [0,1]$.
5. Clip to `[1e-4, 1-1e-4]` to avoid log issues.
6. Write `submission.csv`: columns are `StudyInstanceUID` and the 12 target labels.

Inference speed: ~50–100 test studies per second on a single T4 GPU, so the full test set (if ~2000–5000 studies) completes in 30–60 seconds.

7. Limitations

1. No imaging signal. The model cannot learn visual patterns (bone marrow signal, cartilage thickness, fluid appearance). This ceiling is hard and cannot be overcome with architecture changes alone.
2. Report completeness. If a radiologist overlooked a finding or considered it irrelevant, it will not appear in the report, and the model will not detect it.
3. Report quality and bias. Rare findings may be underreported (radiologists mention effusion every time but only note minor OA when severe). Systematic negation failures or translator errors in multilingual reports will be learned as pattern.
4. Overfitting to the 58 gold labels. The model is trained only on 58 explicit labels; the rest of the training corpus has no supervision. The model may fit spurious correlations in the 58-study subset.
5. Test-set mismatch. If test reports come from a different institution, time period, or language mix than the training reports, performance will degrade.
6. Single-class validation sets. Rare labels may have all positives or all negatives in the validation set, yielding undefined AUC. These classes are skipped in the macro average, potentially masking severe overfitting to one class.

8. Future Improvements

1. Weak supervision from rules. Implement a rule-based labeler (keyword matching, negation handling, laterality resolution) over the unlabeled 4,349 studies to create noisy supervision, then train with noise-robust losses.
2. Ensemble and distillation. Train a smaller model (BERT-base, 110M) and a larger one (XLM-R-large, 340M) and average their predictions, or distill the large model into the small one.
3. Cross-validation. Split the 58 gold labels into 5 folds and train five models, then average their test predictions for more stable estimates.
4. Language-specific models. Train separate models for the top languages (English, Spanish) to avoid dilution from low-resource languages.
5. Data augmentation. Use back-translation (translate to another language and back) or token-level dropout to augment the training set.
6. Prompt tuning or in-context learning. Use a large language model (GPT-3.5, Claude) with few-shot prompts to extract labels from reports, then fine-tune on the resulting pseudo labels.
7. Hybrid multi-modal (if images become available). Concatenate text embeddings with image embeddings and train a fusion model.

9. Conclusion

When images are unavailable, text-only classification of radiology reports is a pragmatic and fast solution. A fine-tuned XLM-RoBERTa-base model reaches a validation macro ROC-AUC of `[[val macro AUC]]` on the 58 explicitly labeled studies, trained in `[[minutes]]` on a single GPU. The main determinant of performance is report quality and completeness rather than model architecture. This baseline is useful as a sanity check and a fallback when images are missing; in production, a multi-modal model (text + image) would be expected to outperform text alone.

References

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